

Assignment

⇒ Type of Data

- ↳ Numerical Data
- ↳ Data represented in no and can be measured

Example Age = 21, Salary = 5000

⇒ Categorical Data

- ↳ Data represented in labels or categories

Example : Gender

⇒ Type of Statistics

- ↳ Descriptive statistics

↳ Summarize and Describe Data

- ↳ Inferential statistics

↳ Makes Prediction or conclusion about a population from a sample

⇒ Measure of Central Tendency

~~Mean~~ Mean (Average)

↪ Sum of all Value Divided by total number of Values

⇒ Median

↪ Middle Value after Sorting Data

⇒ Mode

↪ Most frequently occurring Value

⇒ Measures of Dispersion

↪ Range

↪ Difference between Maximum and Minimum Value

↪ Variance

↪ Measures Spread of Data

$$\text{Variance} = \frac{\sum (x - \bar{x})^2}{n}$$

↪ Standard Deviation

↪ Square root of Variance

$$SD = \sqrt{\text{Variance}}$$

5 $\Rightarrow$

Gaussian Distribution

$\hookrightarrow$ Bell-shaped Symmetric Distribution

6 $\Rightarrow$

log Normal Distribution

$\hookrightarrow$ Data whose logarithm is normal Distributed

7

3-Sigma rule

$\hookrightarrow$ 68% Data within 1 σ

95% within 2 σ

99.7% within 3 σ

 $\Rightarrow$

Quartiles

Divide Data into 4 Parts

$Q_1 = 25\%$

$Q_2 = \text{Median}$

$Q_3 = 75\%$

 $\Rightarrow$

Five Number Summary

$\hookrightarrow$ Minimum

$\hookrightarrow Q_1$

$\hookrightarrow \text{Median}$

$\hookrightarrow Q_3$

$\hookrightarrow \text{Maximum}$

 $\Rightarrow$

Skewness

$\hookrightarrow$ Measure asymmetry of Data

$$= \frac{3(\text{Mean} - \text{Median})}{SD}$$